Biology Course Handbook

Course Code: BIO-11 | Academic Year 2025–2026

Course Description

Introductory Biology exploring cells, genetics, evolution, and ecosystems.

Course Overview

Students investigate living systems from the molecular to the ecosystem scale through labs, dissections, and inquiry-based projects, developing both conceptual understanding and scientific method skills.

Learning Outcomes

By the end of this course, students will be able to:

1. Describe the structure and function of prokaryotic and eukaryotic cells and their organelles.
2. Explain the processes of photosynthesis and cellular respiration and their role in energy flow.
3. Apply the principles of Mendelian genetics to predict inheritance patterns.
4. Explain the theory of evolution by natural selection using supporting evidence.
5. Analyze the flow of energy and cycling of matter within ecosystems.

Assessment

30% labs, 25% tests, 20% project, 25% final exam.